

## Modal Logic: Tableaux Examples

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Example 2:

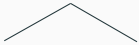
$$(\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B) \text{ in } \mathbf{K}$$

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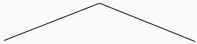
Example 2:  $(\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$  in  $\mathbf{K}$

Let's show this is a theorem in  $\mathbf{K}$ .

1.  $\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$  Assumption
2.  $\mathbb{T} 1, \Diamond A \vee \Diamond B \rightarrow \mathbb{F} 1$
3.  $\mathbb{F} 1, \Diamond(A \vee B) \rightarrow \mathbb{F} 1$

|                                                                                   |                                                                             |                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------|
| 1.                                                                                | $\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ | Assumption                 |
| 2.                                                                                | $\mathbb{T} 1, \Diamond A \vee \Diamond B$                                  | $\rightarrow \mathbb{F} 1$ |
| 3.                                                                                | $\mathbb{F} 1, \Diamond(A \vee B)$                                          | $\rightarrow \mathbb{F} 1$ |
|  |                                                                             |                            |
| 4.                                                                                | $\mathbb{T} 1, \Diamond A \quad \mathbb{T} 1, \Diamond B$                   | $\vee \mathbb{T} 2$        |

The disjunction branches.

|                                                                                   |                                                                             |                            |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------|
| 1.                                                                                | $\mathbb{F} 1, (\Diamond A \vee \Diamond B) \rightarrow \Diamond(A \vee B)$ | Assumption                 |
| 2.                                                                                | $\top 1, \Diamond A \vee \Diamond B$                                        | $\rightarrow \mathbb{F} 1$ |
| 3.                                                                                | $\mathbb{F} 1, \Diamond(A \vee B)$                                          | $\rightarrow \mathbb{F} 1$ |
|  |                                                                             |                            |
| 4.                                                                                | $\top 1, \Diamond A$                                                        | $\vee \top 2$              |
| 5.                                                                                | $\top 1.1, A$                                                               | $\Diamond \top 4$          |
| 6.                                                                                | $\mathbb{F} 1.1, A \vee B$                                                  | $\Diamond \mathbb{F} 3$    |
| 7.                                                                                | $\mathbb{F} 1.1, A$                                                         | $\vee \mathbb{F} 6$        |
| 8.                                                                                | $\otimes$                                                                   | $\vee \mathbb{F} 6$        |
|                                                                                   | $\otimes$                                                                   |                            |

Both branches close, so this **is** valid in K.